

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 36

Title: Glycolysis and Hemolysis (Flipped Classroom Session)

Dr. Clayton Taylor

Email: ctaylor2@sgu.edu

Acknowledgement: Dr. Sharmila Upadhyia and Dr.

Mohamed Idries

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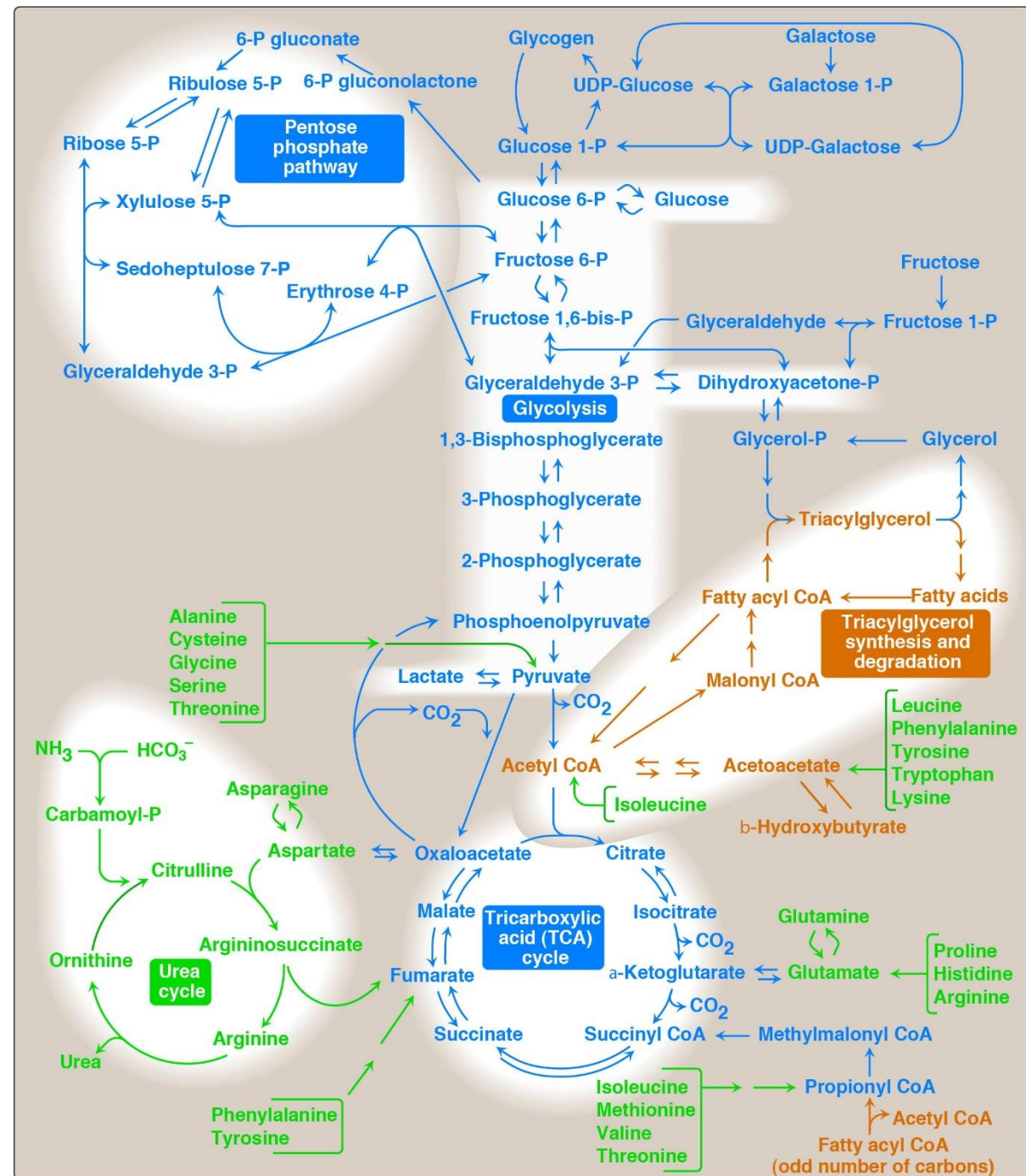
Required reading before Flipped class

- [Introduction to Metabolism and Glycolysis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402)
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- <https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=260211618&bookid=3402#260211666>
- Lippincott's Q: 8.1 -8.8

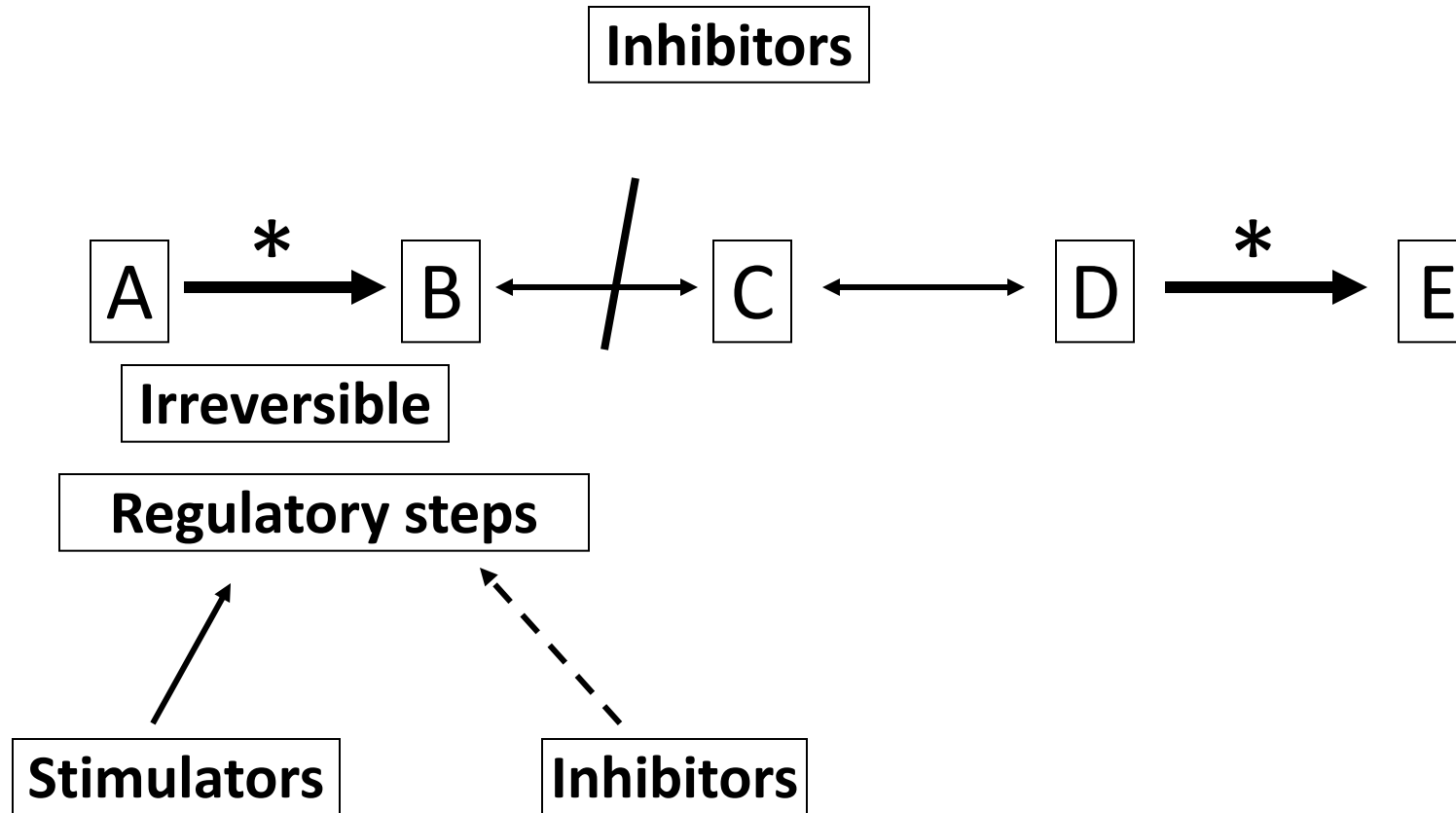
Flipped Class: Glycolysis & Hemolysis

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0298	Define glycolysis and explain its role in the generation of metabolic energy.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0299	List the reactions and enzymes that convert glucose into pyruvate (emphasize key enzymes and irreversible reactions)
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0300	Define substrate level phosphorylation and identify those reactions in glycolytic pathway
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0301	Explain significance of lactate production in anaerobic glycolysis and describe the fate of lactate formed in RBC and muscle (Cori cycle)
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0302	Outline the regulation of glycolysis indicating the regulatory enzymes and highlight the role of AMP, ATP on glycolysis.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0303	Review the significance of glucose transporters (GLUT1-5) present in various tissues
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0304	Compare and contrast the action of glucokinase and hexokinase (with the help of a graph)
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0305	Review the regulation of glycolysis
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0306	Outline the function of glycolysis in specific tissues (RBC) including 2,3-BPG formation and energy
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0307	Analyze the effect of fluoride ions on glycolytic enzymes
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0308	Identify the biochemical mechanisms involved in the occurrence of lactic acidosis in states of poor tissue perfusion/ oxygenation.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0309	Explain the basis for occurrence of hemolysis in inherited pyruvate kinase deficiency
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0310	Compare the changes in 2,3 BPG levels and hemoglobin oxygen dissociation curve in hexokinase and pyruvate kinase deficiency
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0311	Differentiate between pyruvate kinase deficiency (defect in glycolytic pathway) and glucose 6-phosphate dehydrogenase deficiency (pentose phosphate defect)

Intermediary metabolism



Metabolic pathways



Clicker question NEXT!!!

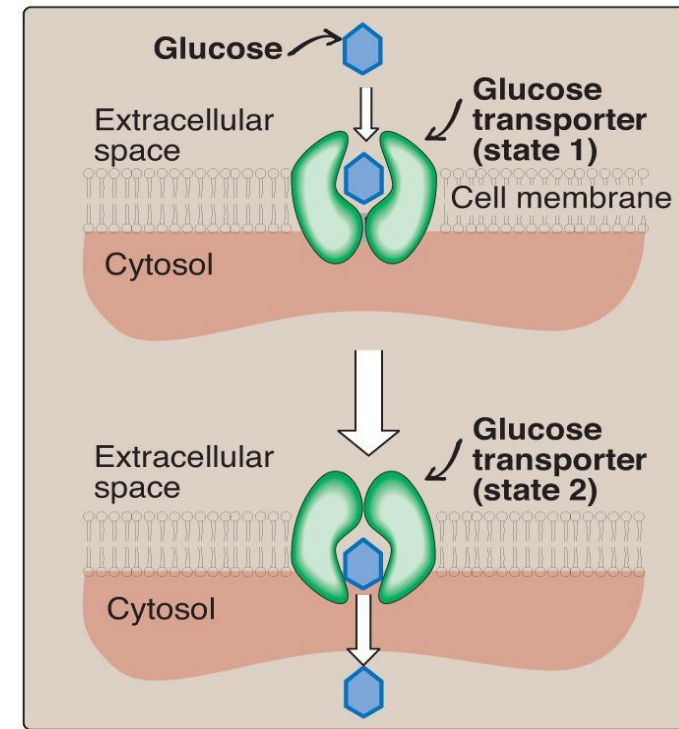
Clicker Question

An investigator is studying the entry of glucose into the red blood cells. Which of the following is the most likely mechanism of transport into these cells?

- A. Secondary active transport 20%
- B. Primary active transport 20%
- ✓ C. Facilitated diffusion 20%
- D. Ion channels 20%
- E. ABC transporter 20%

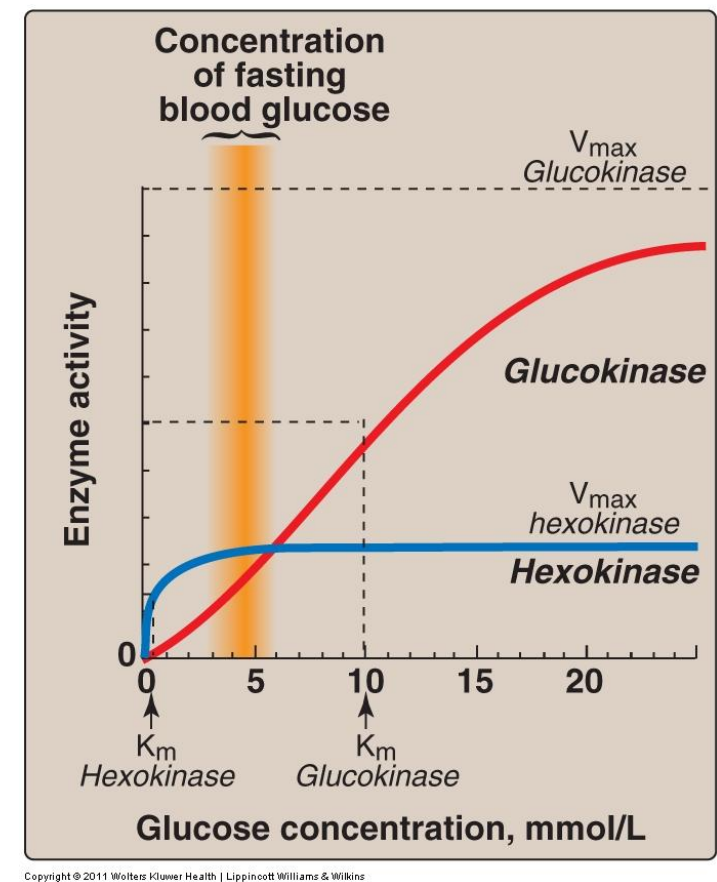
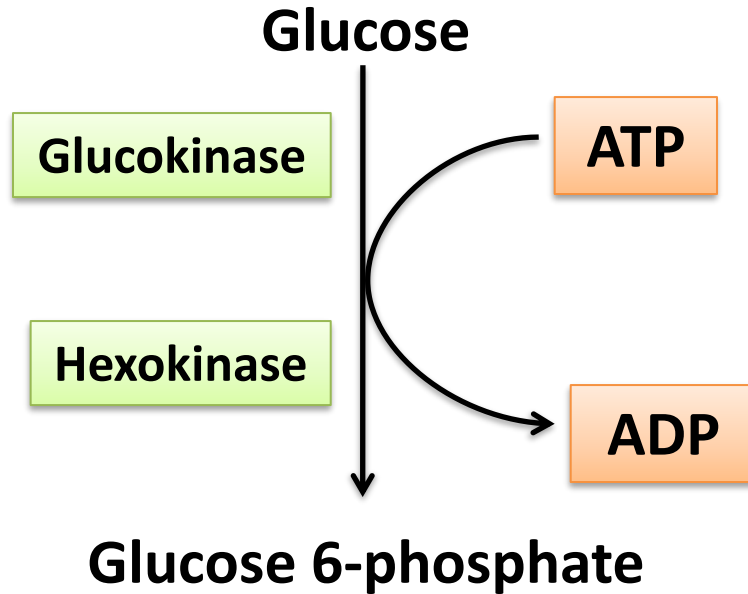
Fate of absorbed glucose

- Glucose from the bloodstream enters cells via specific transporters (GLUT) by **facilitated diffusion**
 - GLUT 2 – liver
 - GLUT 1 and 3 – in neurons and brain
 - **GLUT 1 in erythrocytes**
 - GLUT 4 in adipose tissue and muscle; Responsive to insulin. Number of GLUT4 increase in presence of insulin, when the plasma glucose level is elevated.



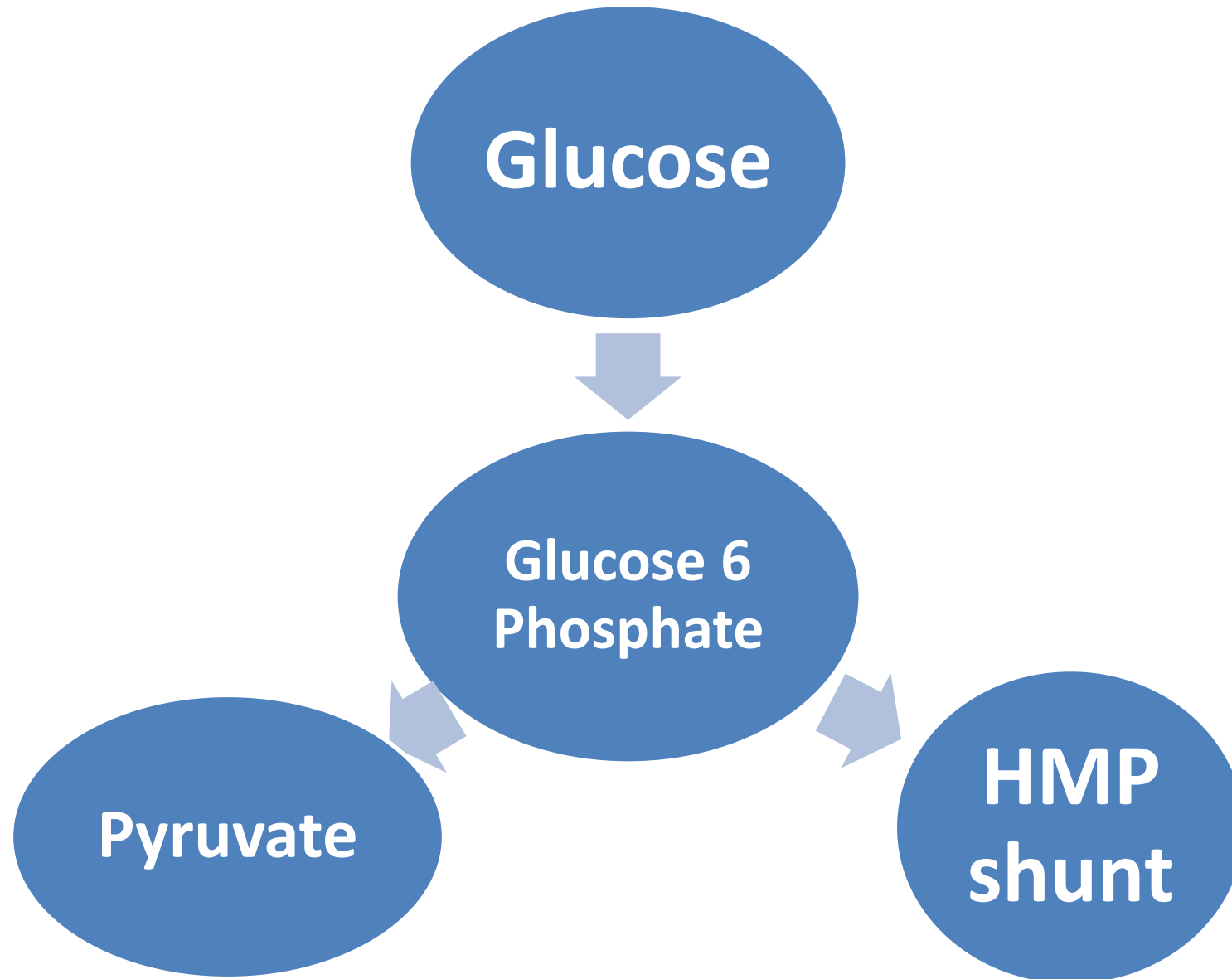
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Formation of glucose 6-phosphate



- Glucose is phosphorylated to glucose 6-phosphate
 - Glucokinase: Liver has high K_m for glucose (more active when plasma glucose is elevated). Also present in β -cells of pancreas.
 - Other tissues (RBC) – contain **hexokinase**; Low K_m for glucose

Intracellular fates of glucose 6-phosphate in RBC



Clicker Question

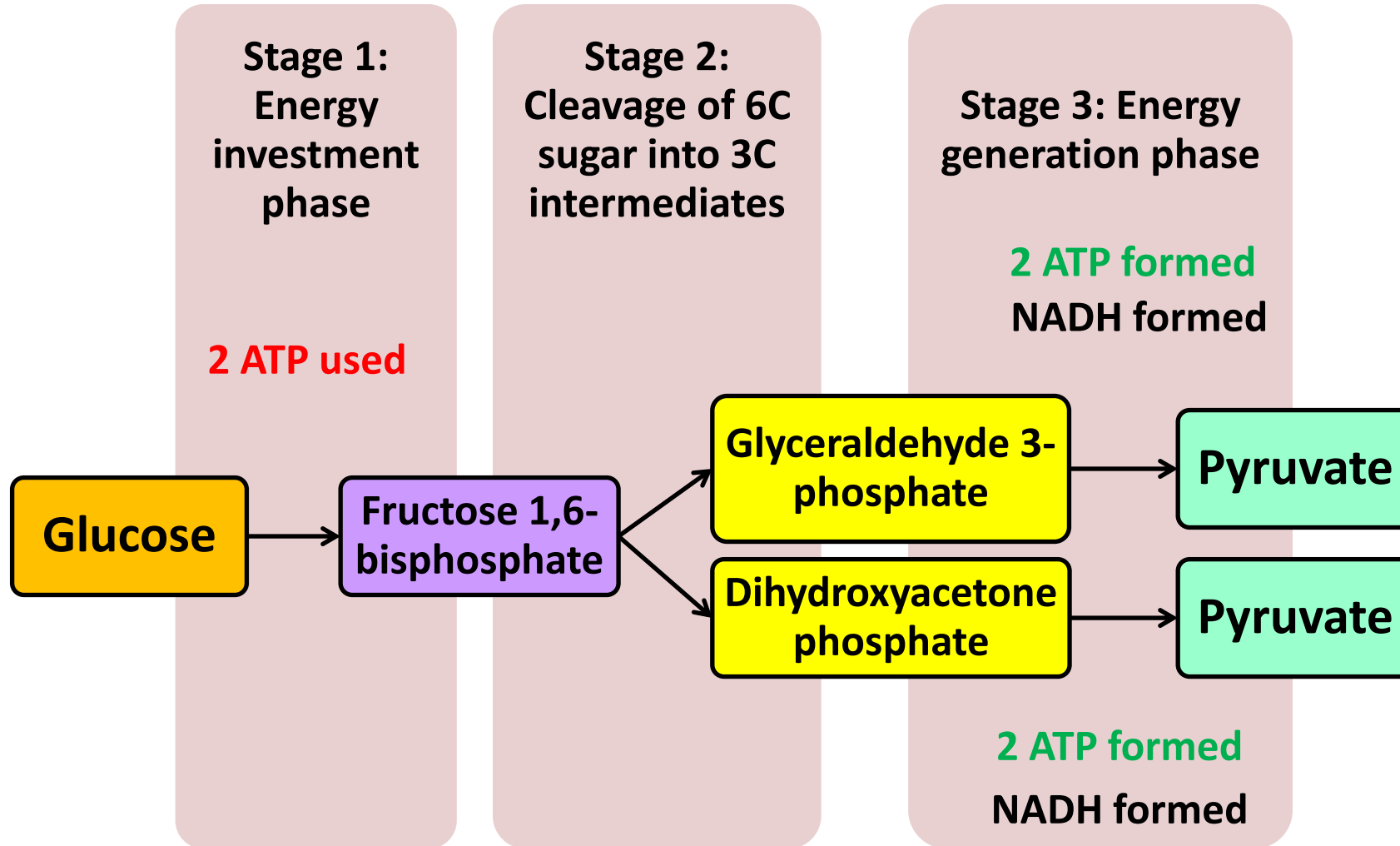
A researcher is studying the pathway which converts glucose to lactate in mature red blood cells. Which of the following best describes this pathway?

- ✓ A. Forms ATP 0%
- B. Forms NADPH 0%
- C. Generates ribose 5-phosphate 0%
- D. Occurs in mitochondria 0%
- E. Requires oxygen 0%

Glycolysis

- Principal pathways for generating ATP (energy)
 - In brain and **erythrocytes**
- **Generates ATP** in presence and absence of O₂ (aerobic and anaerobic) and in presence and absence of mitochondria
- Red blood cells have **no mitochondria**
 - Glycolysis is the only source of ATP in these cells
- **Glycolytic enzymes** are present in the **cytosol**
- Remember: Enzymes of **pentose phosphate pathway** are also present in **cytosol**

Overview of phases of glycolysis



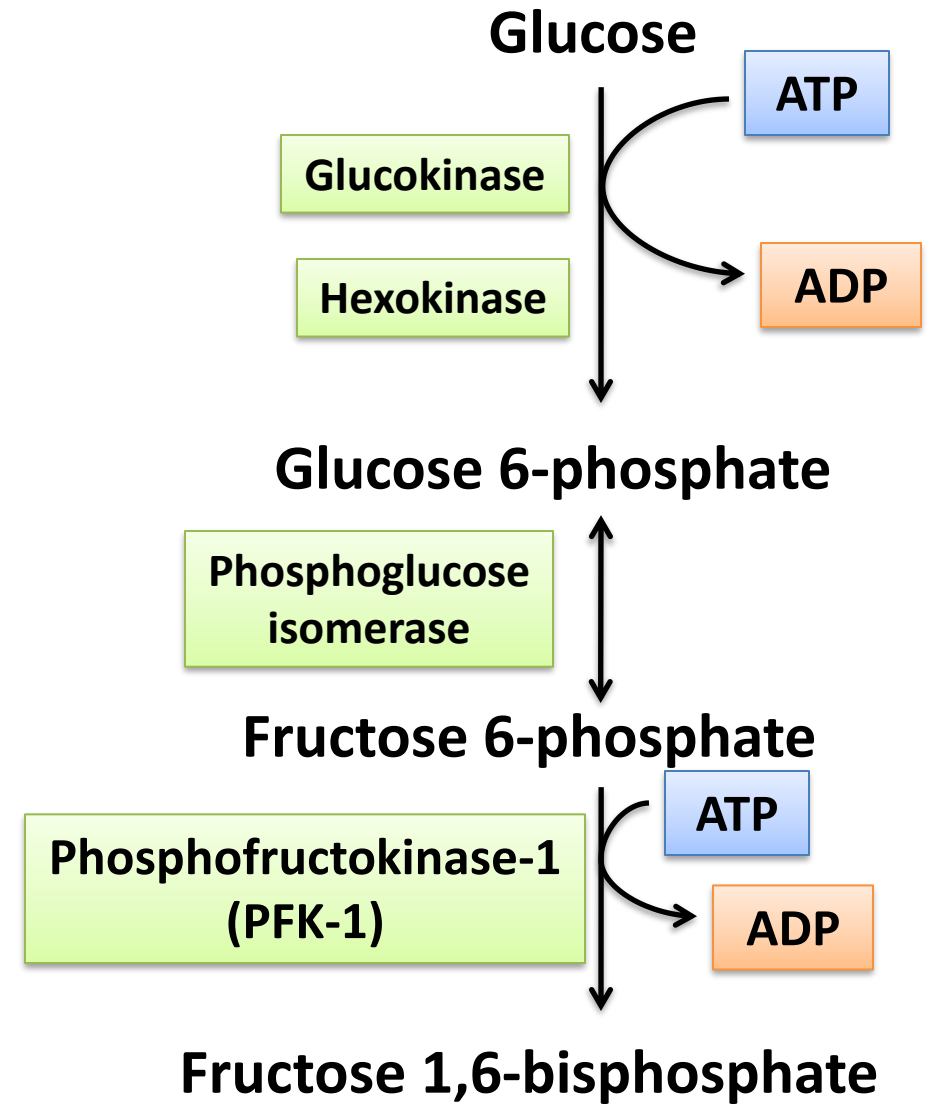
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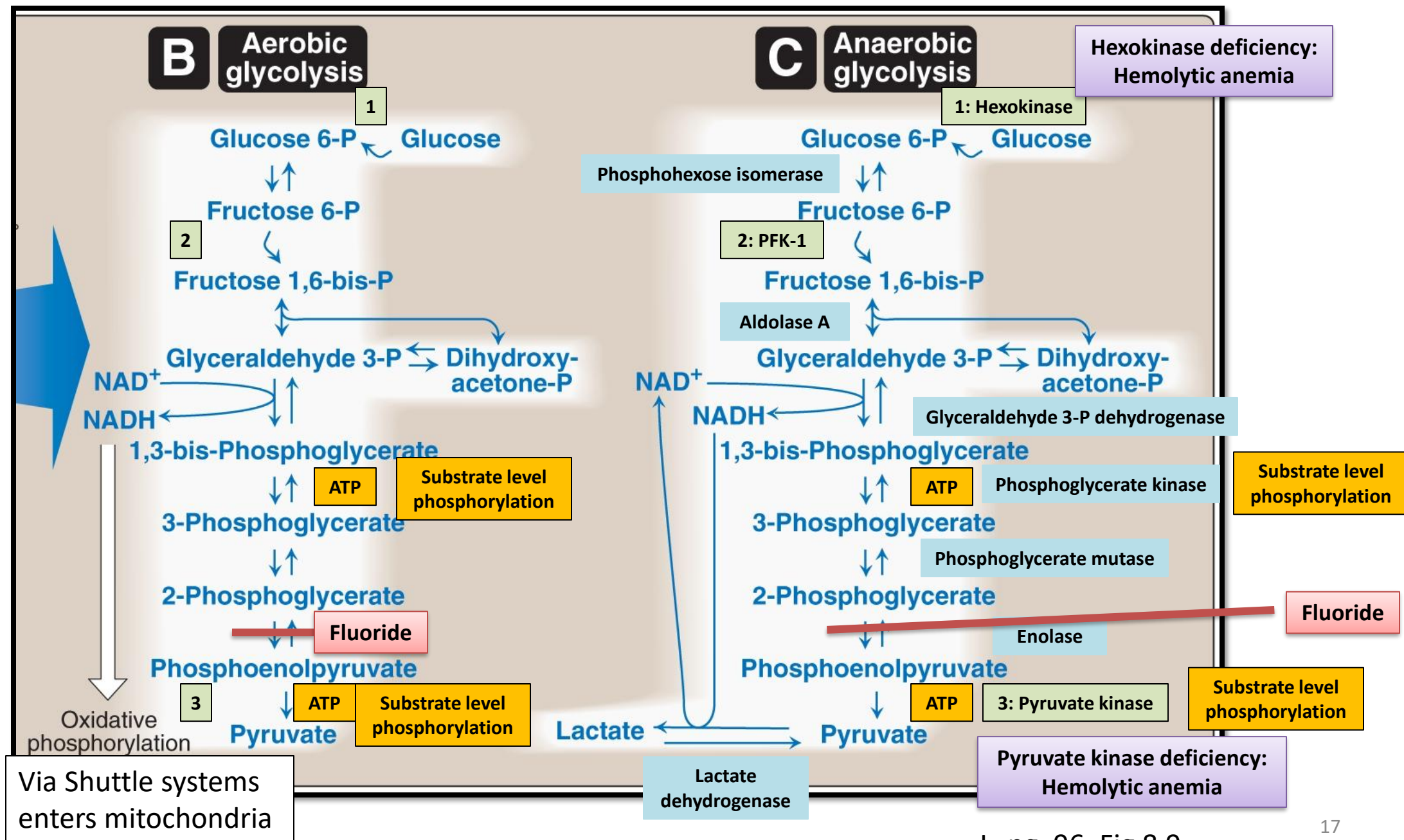
Which enzyme/s directly forms ATP for RBC metabolism?

- A. Lactate dehydrogenase 0%
- B. Pyruvate dehydrogenase 0%
- ✓ C. Phosphoglycerate kinase 0%
- D. Glyceraldehyde 3-phosphate dehydrogenase 0%
- E. Glucose 6-phosphate dehydrogenase 0%
- ✓ F. Pyruvate kinase 0%
- G. Phosphofructokinase-1 0%
- H. Hexokinase 0%

Irreversible/ Regulated reactions

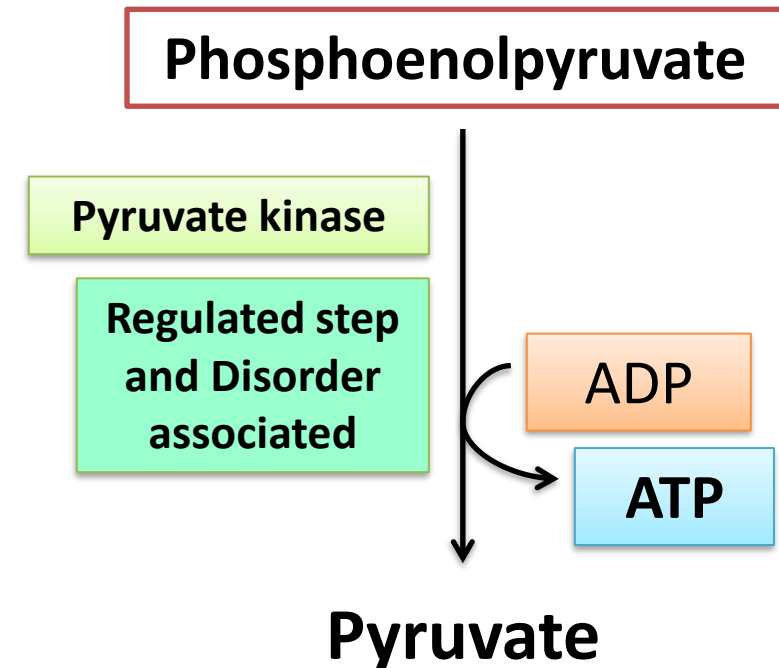
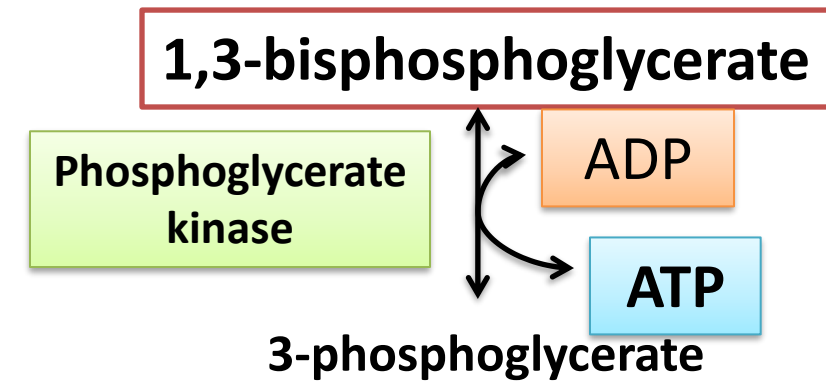
- Two phosphorylation reactions require ATP (energy investment phase) – requires 2 ATP
- **Glucokinase/ Hexokinase**
- **PFK-1** : Irreversible reaction of glycolysis; Regulated step (**allosteric regulated** enzyme)
- **Pyruvate kinase** is also irreversible





Energy Generation: Substrate level phosphorylation reactions

- Reactions directly form ATP from ADP
- DO NOT require the Electron Transport chain or mitochondria or Oxygen
- ONLY reactions that form ATP in cells that lack mitochondria and in hypoxic conditions
- Forms ATP directly from high energy compounds (1,3-Bisphosphoglycerate and Phosphoenolpyruvate)

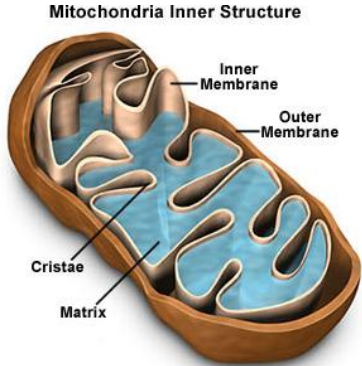


Clicker Question

A researcher is studying lactate formation. Which of the following best describes formation of this intermediate?

- A. Formed by pyruvate dehydrogenase 0%
- ✓ B. Favored when NADH levels are high 0%
- C. Formed in the mitochondria 0%
- D. Lactate dehydrogenase catalyzes an irreversible reaction 0%
- E. Normally formed in the cardiac muscle cells 0%

Fates of pyruvate



Pyruvate

Aerobic (abundance of oxygen and mitochondria)

Anaerobic (no mitochondria OR no oxygen)

Absence of mitochondria (red blood cells, lens, cornea)
Poorly vascularized tissues or lack of oxygen (actively contracting white skeletal muscle, cardiac muscle hypoxia) to lactate (lactic acid)

Acetyl CoA
(Pyruvate dehydrogenase)

Lactate (Lactate dehydrogenase)

Cytosolic enzyme

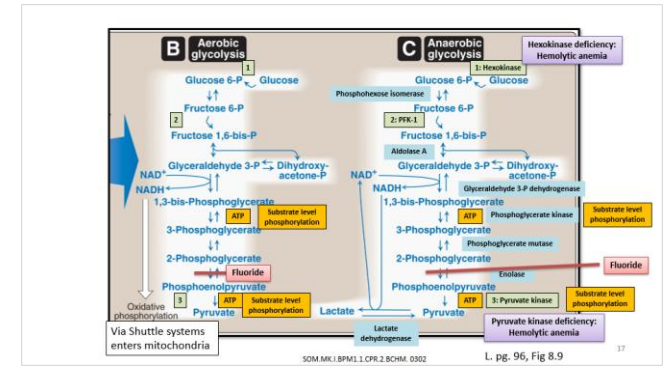
Enters TCA cycle

Goes to liver via Cori cycle

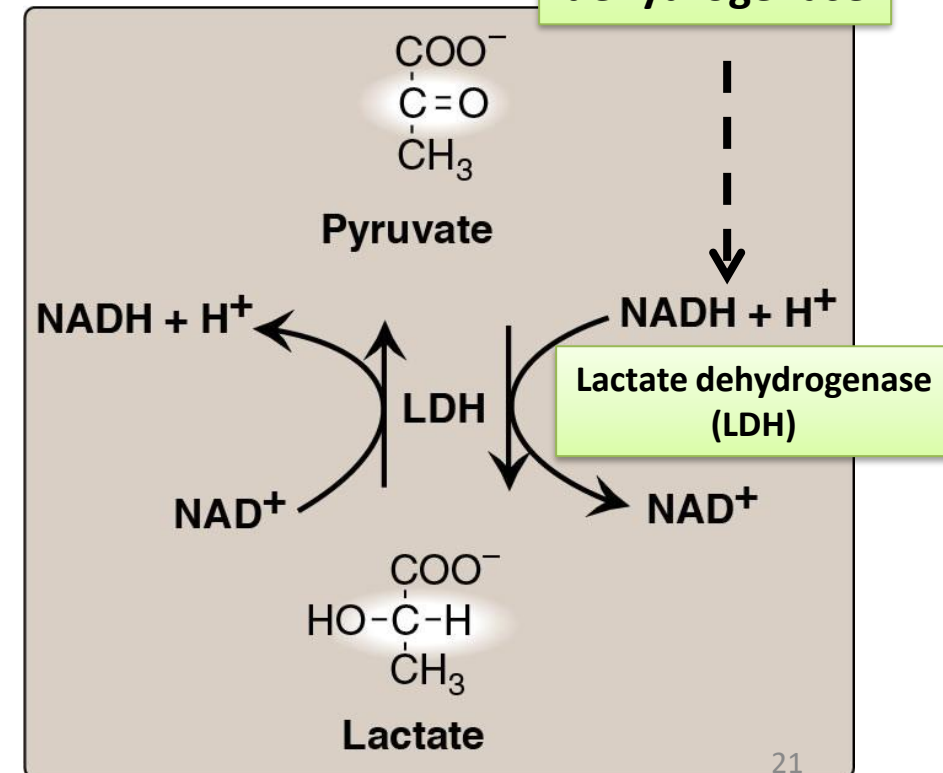
Mitochondrial enzyme

Anaerobic fate of pyruvate

- NADH formed by glyceraldehyde 3-phosphate dehydrogenase reaction is reoxidized to form NAD^+
 - Allows glycolysis to proceed even in absence of mitochondria
- LDH is **cytosolic enzyme**
- Direction of reaction depends on **$\text{NADH} / \text{NAD}^+$ ratio**.
 - Higher NADH levels, favors lactate formation
- Identify states that result in **high $\text{NADH} / \text{NAD}^+$ ratio**



Glyceraldehyde 3-phosphate dehydrogenase

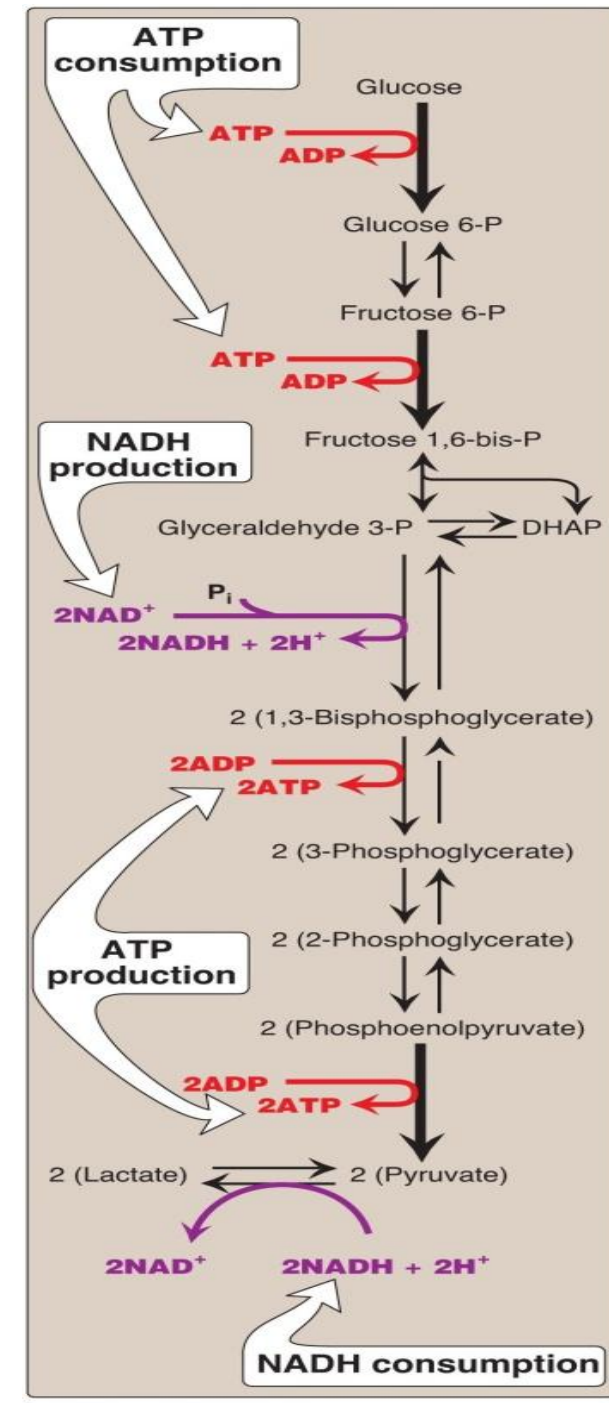


Lactate formation by anaerobic glycolysis in muscle

- In actively contracting skeletal muscle, NADH formed as a result of glycolysis and TCA cycle, exceeds oxidative capacity of ETC, resulting in a **high NADH/NAD⁺ ratio**.
 - Favors lactate formation via lactate dehydrogenase
 - Lactate (lactic acid) in muscle causes a drop in pH and causes **cramps**
- Cardiac muscle has low NADH/ NAD⁺ ratio and is generally aerobic (Abundant mitochondria and good vascularization).
 - Converts lactate to pyruvate and later to acetyl CoA, to enter TCA cycle
 - Cardiac muscle hypoxia results in lactate formation (myocardial infarction)

Summary: Key features of glycolysis

- Oxidation of glucose to 2 molecules of pyruvate
- Cytosolic pathway
- Three irreversible reactions (regulatory steps)
 - Glucokinase/ hexokinase
 - **Phosphofructokinase-1**
 - Pyruvate kinase
- Two **substrate level phosphorylation** reactions (**ATP formed**)
 - Phosphoglycerate kinase
 - Pyruvate kinase
- Fate of pyruvate
 - Aerobic: Pyruvate transported to **mitochondria** and converted to acetyl CoA (pyruvate dehydrogenase)
 - Anaerobic: Pyruvate converted to lactate in **cytosol** (lactate dehydrogenase)



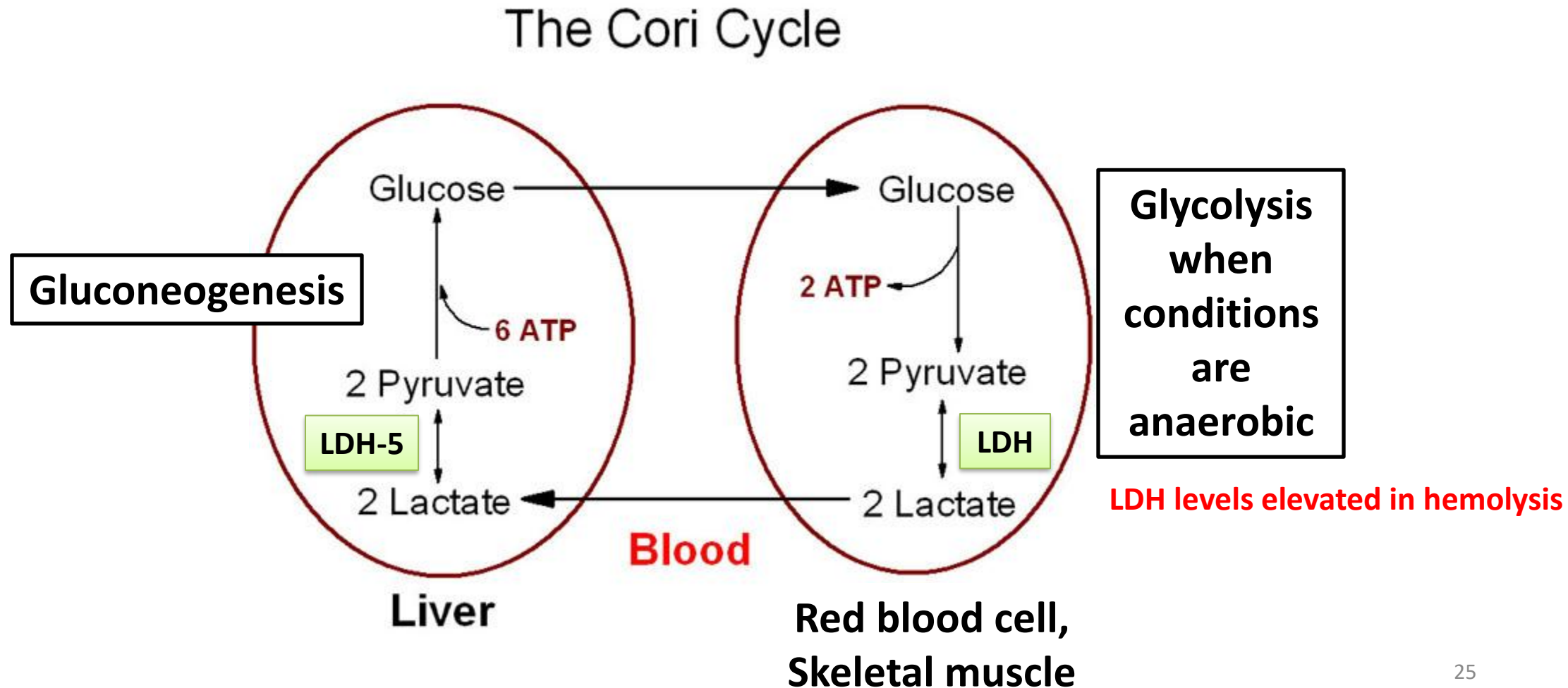
Clicker Question

Which of the following best describes the Cori cycle?

- A. Lactate is formed in the liver 0%
- B. Lactate is formed in the brain 0%
- ✓ C. Lactate is formed in actively contracting skeletal muscle 0%
- D. Lactate is converted to glucose in the heart 0%
- E. Lactate formed in the liver is transported to the intestine 0%

Fate of lactate: Cori cycle

Lactate from muscle and red blood cells, diffuses out of these cells and is taken to liver where it is used for gluconeogenesis (Cori cycle)



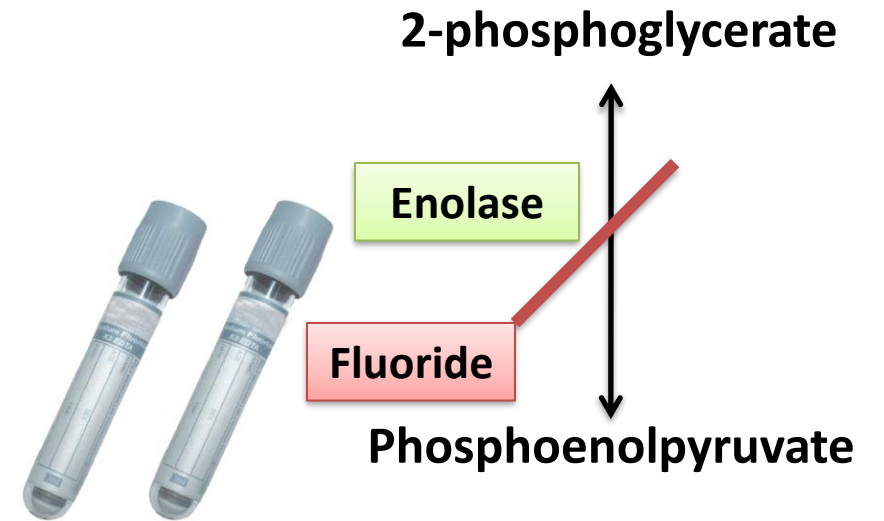
Energetics of glycolysis under anaerobic conditions

(Lack of oxygen/ no mitochondria)

- Overall reaction (anaerobic)
- Glucose → 2 mols of Lactate
- 2 NADH + H⁺ (formed in glyceraldehyde 3-phosphate dehydrogenase reaction) are used in lactate dehydrogenase reaction
- 4 ATP (formed in **substrate level phosphorylation** reactions) (2ATP from each mol of glyceraldehyde 3 phosphate)
- -2ATP used in the energy investment phase
- Total = 2 ATP (Anaerobic glycolysis)
- **Only source of ATP in cells with no mitochondria or hypoxia**

Inhibitors of glycolysis

- In clinical lab, blood collected in **fluoride** containing tubes for estimation of plasma glucose
- If fluoride is not added, plasma glucose is used by RBC and WBC, resulting in erroneously low values
- Fluoride inhibits **Enolase (glycolysis)**

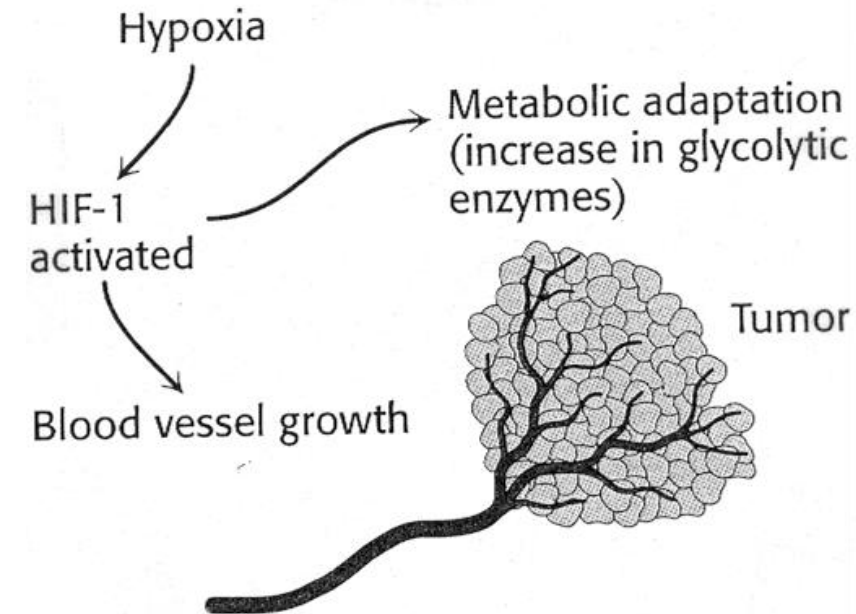


Significance of glycolysis

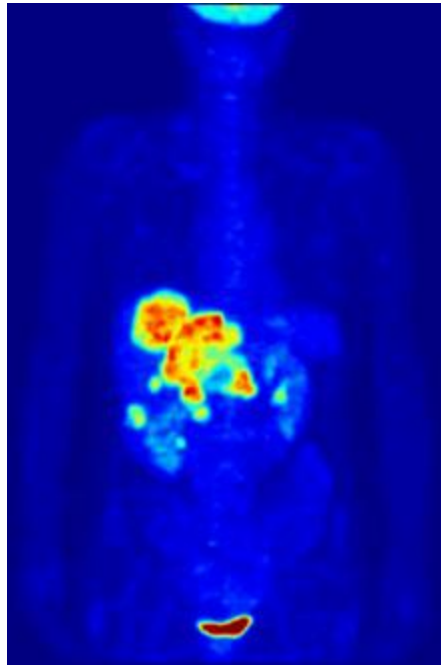
- Tissues in which complete oxidation of glucose takes place (brain, skeletal muscle), glucose is converted to pyruvate, later converted to acetyl CoA, that enters TCA cycle for complete oxidation of glucose
- Tissues of eye (lack mitochondria): cornea, lens
- Tumor tissue uses glycolysis
- **In red blood cells, glycolysis (anaerobic conditions)**
 - Major source of energy (ATP)
 - Forms **2,3 bisphosphoglycerate** (2,3 BPG)

Metabolic changes in cancer cells

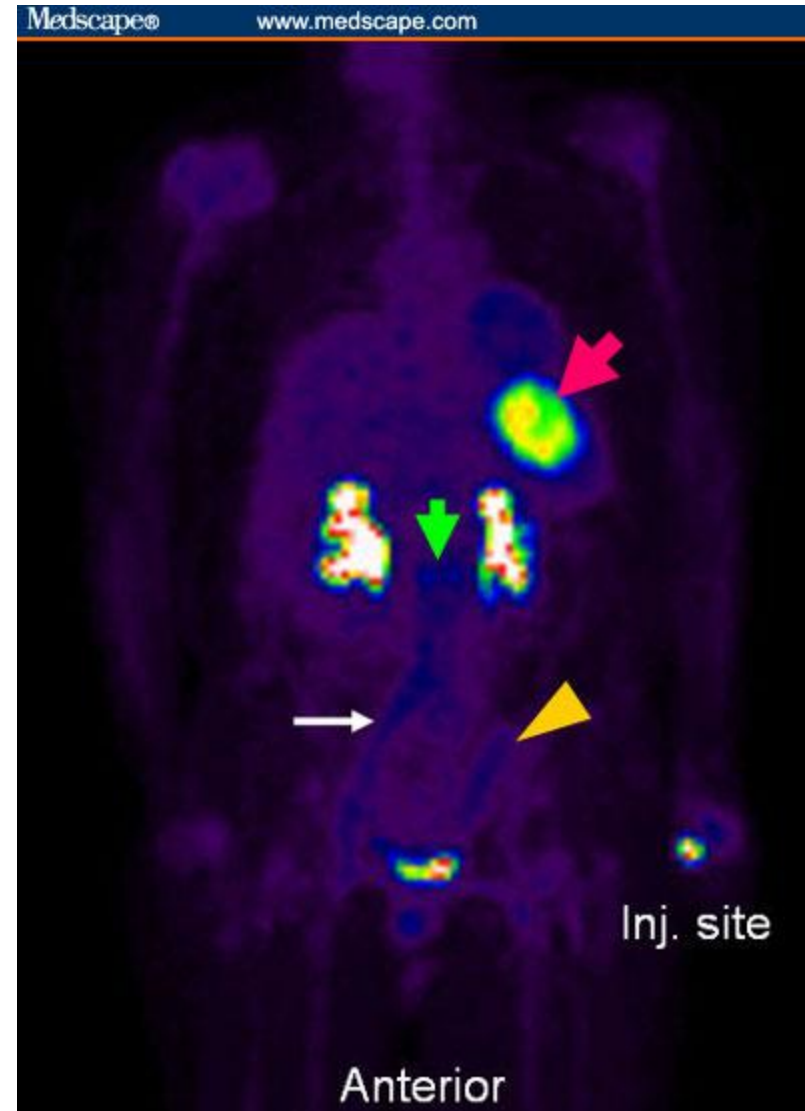
- Cancer cells use glycolysis as source of ATP (energy) – '**Warburg effect**'
- Glucose analog 'fluorodeoxyglucose (FDG)' used in positron emission tomography (PET)
 - Increased uptake and utilization of FDG by tumor
- Inhibitors of glycolysis are explored for use as therapeutic agents in cancers



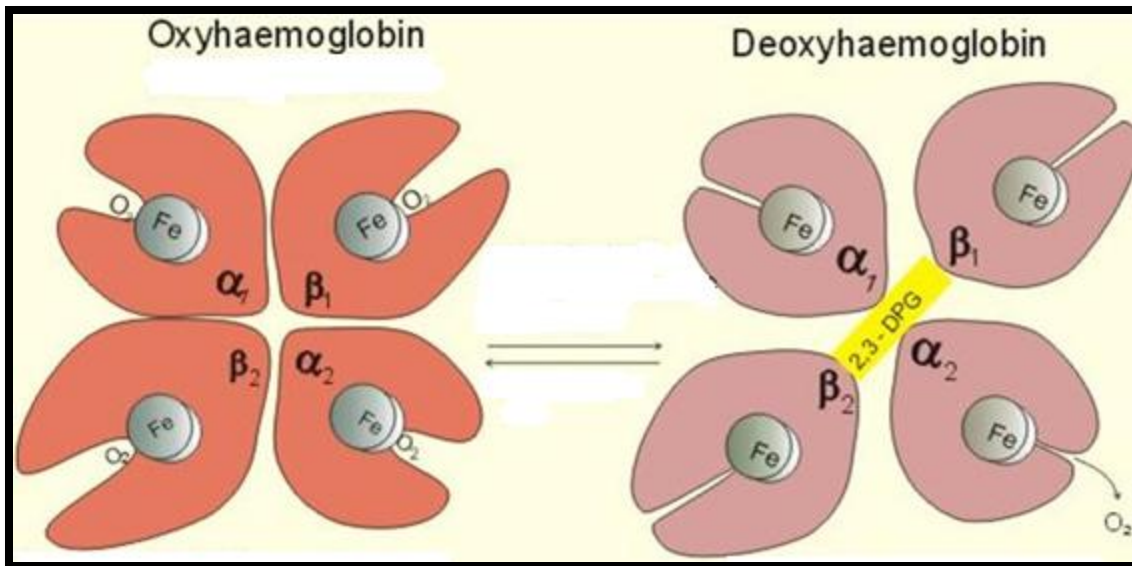
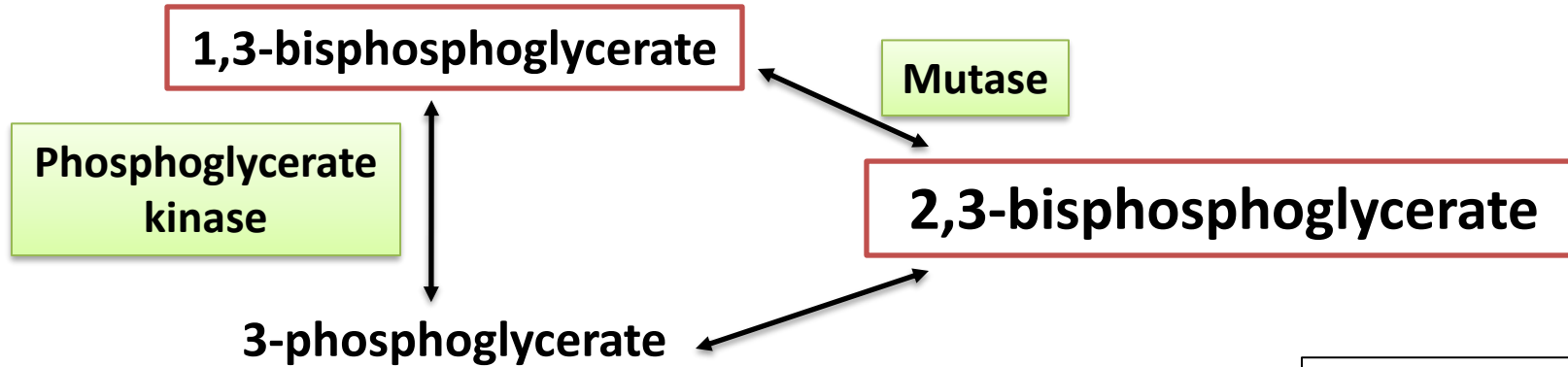
Positron Emission Tomography (PET) scan using fluorodeoxyglucose



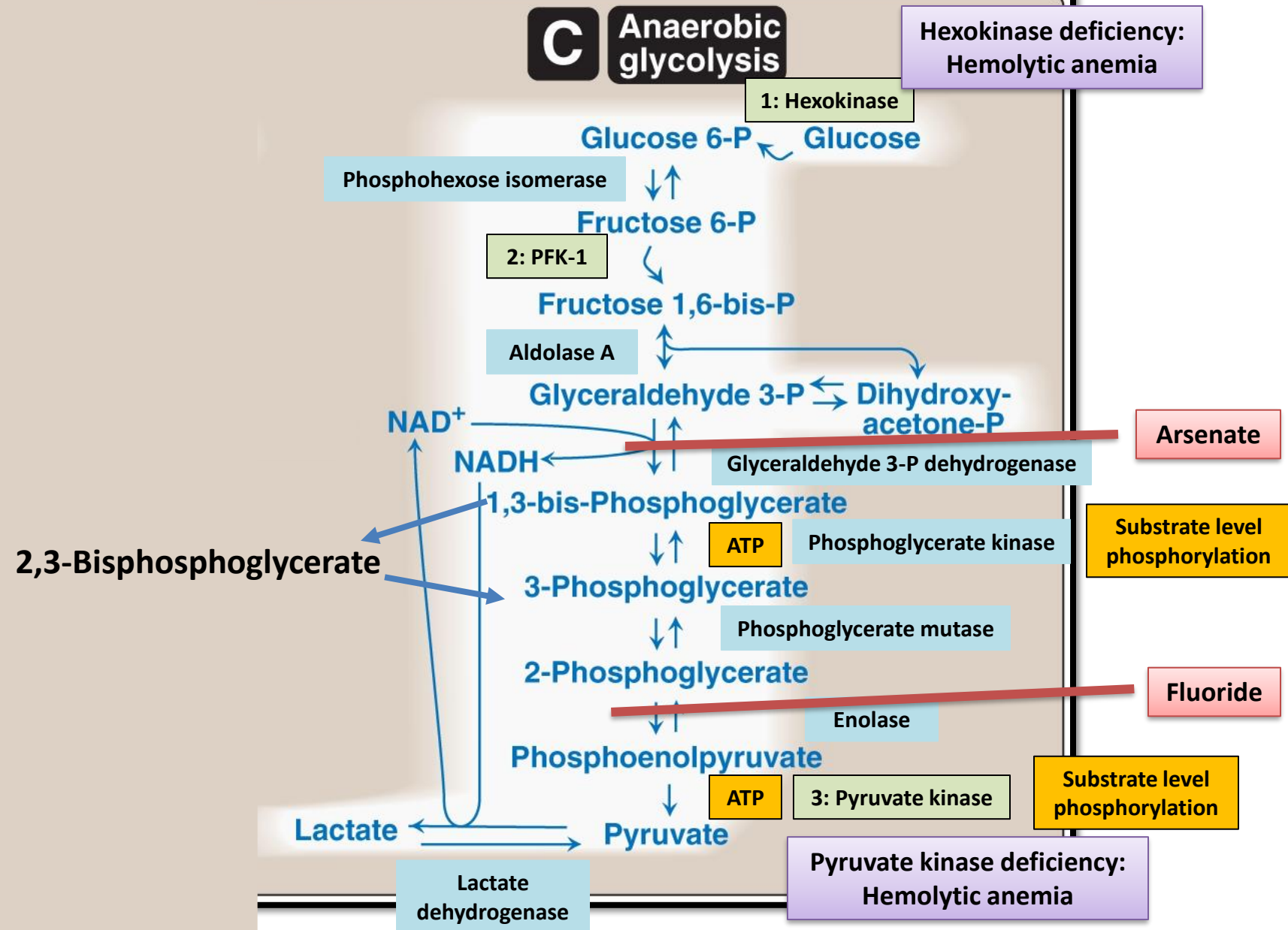
Increased uptake and utilization of FDG by tumor



Significance of glycolysis in RBC: **2,3-BPG formation**



- 2,3-BPG binds to hemoglobin and helps to unload oxygen in tissues
- **2,3-BPG levels are increased** at high altitude
- Decreases Hb oxygen affinity (refer: Hemoglobin lecture)

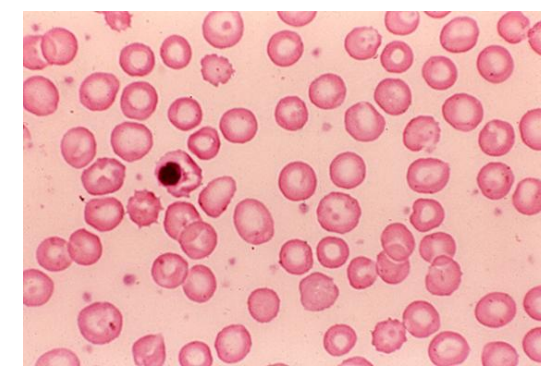
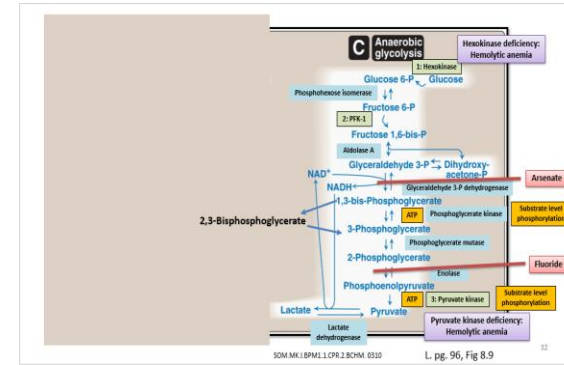


Significance of glycolysis in RBCs: Energy (ATP)

- Glycolysis is the only source of energy (ATP) for RBCs
- Mature RBCs lack mitochondria and are completely dependent on glycolysis for ATP formation
- ATP for RBCs is required for maintenance of Na⁺-K⁺ ATPase (membrane pumps) and maintenance of osmotic balance within RBCs
- Hexokinase or Pyruvate kinase deficiency result in hemolytic anemia

Hemolytic anemia

- Erythrocyte pyruvate kinase deficiency
- ATP required in RBCs to maintain electrolyte concentration (**operation of $\text{Na}^+\text{-K}^+$ ATPase/ membrane pumps**).
- Glycolytic defects cause hemolysis and hemolytic anemia: **Decreased ATP generation and osmotic imbalance**
- Pyruvate kinase deficiency: second most common form of hemolytic anemia due to RBC enzyme deficiency (most common: G6PD deficiency).
- **Lab finding: Serum lactate dehydrogenase is elevated** (release of enzyme from hemolyzed RBCs)
- Compare 2,3-BPG levels in Pyruvate kinase and Hexokinase deficient RBCs (Refer: Hemoglobin lecture) and the oxygen dissociation curve
- Differentiate between G6PD and pyruvate kinase deficiency



Mechanism of hemolysis in pyruvate kinase/ hexokinase deficiency

Defect in Glycolysis (pyruvate kinase/ hexokinase deficiency)



Reduced energy (ATP) generation



Erythrocytes are energy (ATP) deficient



Na^+/K^+ ATPase is compromised (Remember: 40-50% energy in resting cells is used for maintenance of Na^+/K^+ ATPase or membrane pumps)

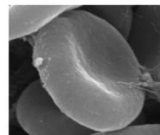


Osmotic imbalance due to ionic imbalance (Na^+ content in affected RBCs higher than in normal cells)



RBC lysis

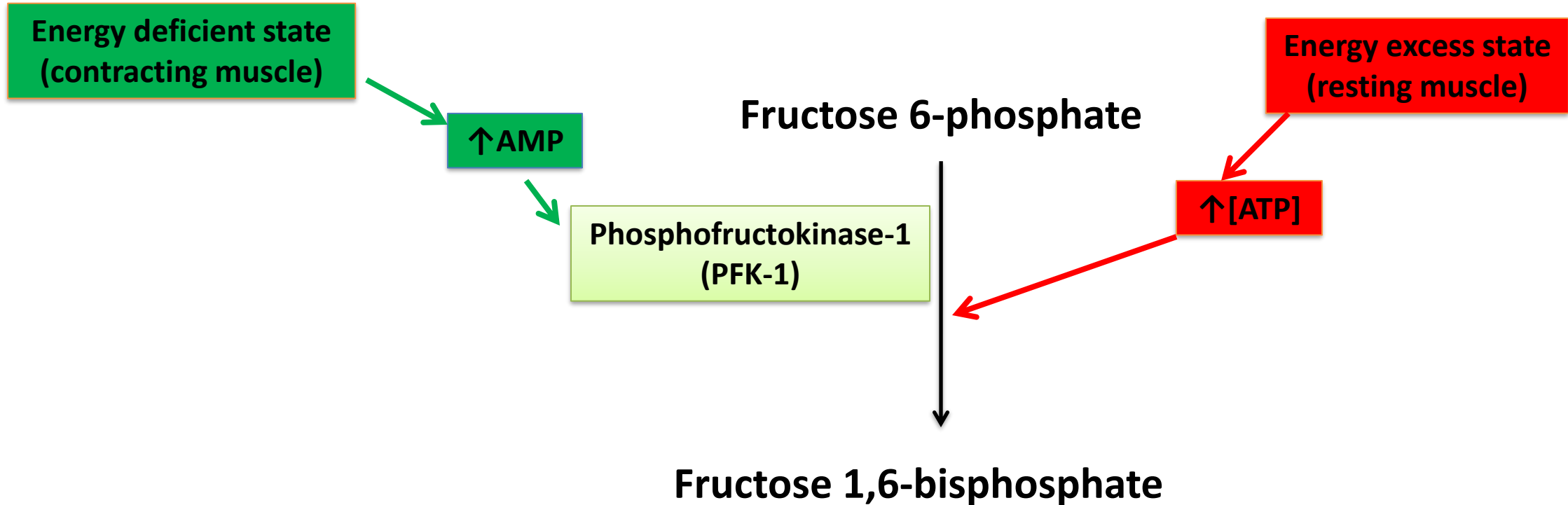
Normal RBC



Lactic acidosis

- Lactic acidosis causes **metabolic acidosis**
 - Weak acid and gives off protons (H^+). H^+ ions buffered by bicarbonate (HCO_3^-); HCO_3^- used up for buffering H^+ from lactic acid
 - pH is low; serum HCO_3^- is decreased; PCO_2 is decreased as a compensatory response; Results in **high anion gap acidosis**
- Lactic acidosis is observed when
 - Increased $NADH/NAD^+$ ratio: Indicates anaerobic respiration
 - Inherited deficiency of pyruvate dehydrogenase (Impaired aerobic metabolism)
 - **Decreased blood supply** due to cardiac failure: **Anaerobic metabolism** in peripheral tissues. **Defect in Cori cycle** and lactate is not efficiently transported to liver.
 - **Pulmonary embolism** (poor oxygenation)
 - **Hemorrhage and circulatory shock**

Regulation of glycolysis in muscle



- PFK-1 : Allosterically regulated enzyme
- Regulation in muscle:
 - High levels of ATP (inhibits glycolysis)
 - & high levels of AMP (stimulates glycolysis)

Clicker Question

Which of the following conditions may result in an increase in lactate levels in the blood?

- ✓ A. Tissue hypoxia 0%
- B. High arterial PO₂ levels 0%
- ✓ C. High NADH/NAD⁺ ratio 0%
- ✓ D. Strenuous muscle activity 0%
- ✓ E. Decreased blood circulation 0%

Clicker Question

A 10-year-old boy is brought to the physician by his parents because of a 4-month history of pallor. Laboratory studies show hemolytic anemia due to pyruvate kinase deficiency. Which of the following best explains the cause of hemolysis in this patient?

- A. Increased 2,3 BPG levels 0%
- ✓ B. Decreased ATP generation 0%
- C. Increased anaerobic glycolysis 0%
- D. Decreased cAMP generation 0%
- E. Increased accumulation of glucose 6-phosphate 0%
- F. Increased reactive oxygen species in RBC 0%

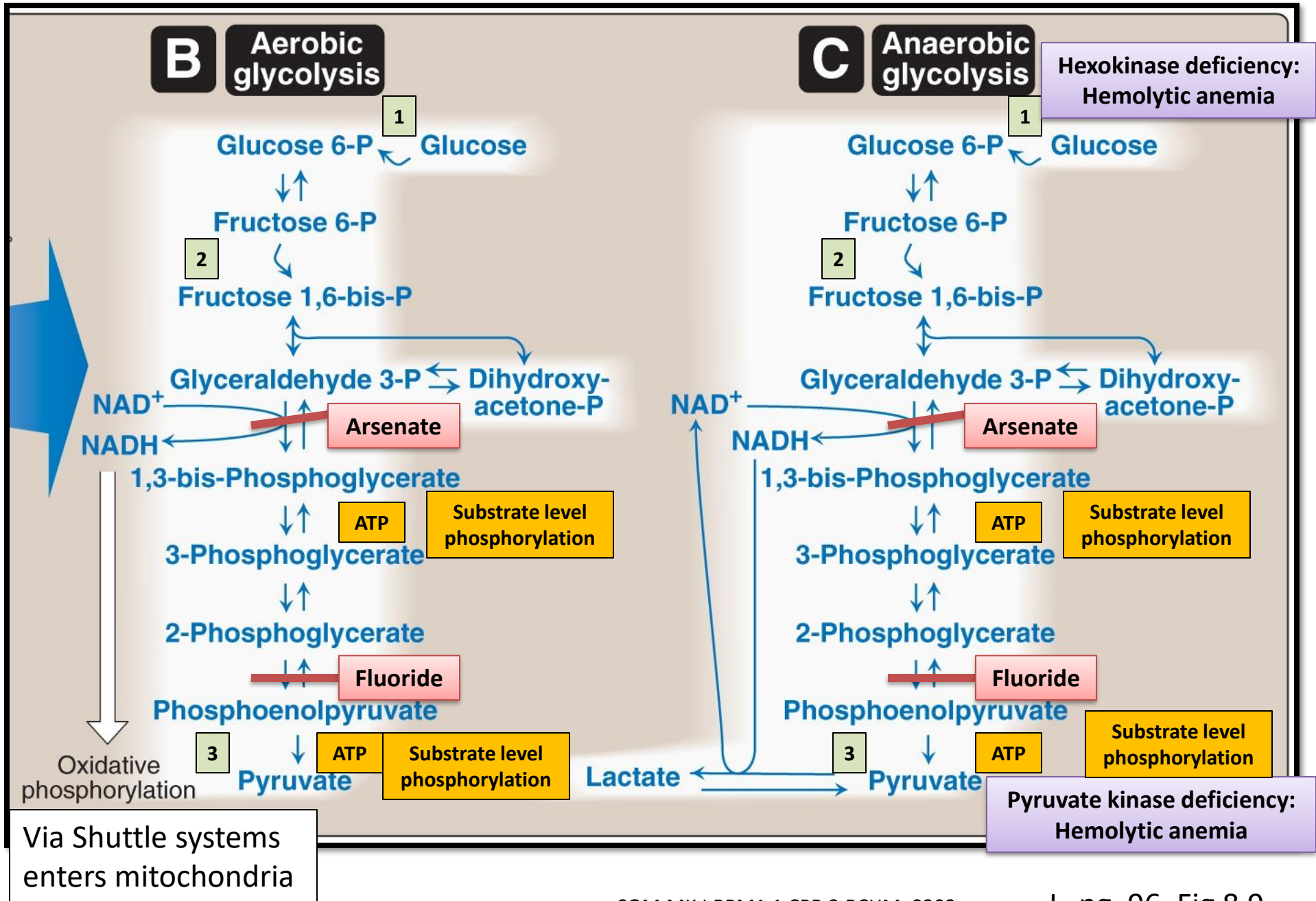
Review the mechanism of hemolysis in G6PD deficiency and pyruvate kinase and hexokinase deficiency

Clicker Question

A researcher is studying the conversion of lactate to glucose in the liver. Which of the following enzymes are NOT used in the conversion of lactate to glucose?

- | | |
|---|----|
| A. Phosphoglycerate kinase | 0% |
| B. Glyceraldehyde 3-phosphate dehydrogenase | 0% |
| ✓ C. Glucokinase | 0% |
| ✓ D. Phosphofructokinase-1 | 0% |
| E. Lactate dehydrogenase | 0% |
| ✓ F. Pyruvate kinase | 0% |

Identify the irreversible reactions of glycolysis



Concept map

